

Md Kaif Nezami

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Profile

Second-year ECE student at BIT Sindri with hands-on experience in UAV electronics, embedded systems, and sensor integration. Currently contributing to ISRO IROC 2026 UAV docking work through subsystem design, component selection, and testing. Seeking a research internship in UAV development and autonomous aerial systems.

Education

Bachelor of Technology in Electronics and Communication Engineering **2024 – 2028**
Birsa Institute of Technology (BIT) Sindri, Dhanbad, Jharkhand CGPA: 7.10 (1st Sem)

Skills

- **Embedded Systems:** Microcontrollers, sensor interfacing, communication modules
- **UAV Systems:** Mission planning, flight testing, troubleshooting, autonomous systems basics
- **Electronics:** PCB basics, UAV power systems, ESC integration, circuit design
- **Sensors:** IMU, ultrasonic, proximity sensor integration
- **Programming:** C, Python
- **Tools:** Git, GitHub, Visual Studio Code, Linux CLI, Google Colab

Soft Skills: Problem-solving · Team collaboration · Communication · Attention to detail

Projects

ISRO IROC 2026 – Autonomous UAV Docking Challenge *Jan 2026 – Present*

Qualified Round 1 and Round 2; currently in Round 3 – full system autonomy phase. Leading docking station design improvement as part of hardware implementation.

- Spearheading docking station redesign to improve mechanical precision and reliable UAV docking alignment.
- Integrated proximity, IMU, and ultrasonic sensors for precision landing and autonomous docking operations.
- Performed component selection, circuit design, system integration, and hardware troubleshooting.

Custom Drone – IIT ISM Bootcamp *Jun 2024*

- Assembled a hexacopter using OrangeCube flight controller during a hands-on drone bootcamp at IIT ISM.
- Performed component integration, PID tuning, and autonomous waypoint mission planning.

UAV Payload Setup – Camera Module & Wireless Transmission *2024*

- Worked on a small UAV payload setup integrating a camera module with wireless data transmission.
- Handled wiring, module mounting, and basic signal transmission testing on a UAV frame.

Achievements

- Qualified Round 1 & 2 of **ISRO IROC 2026**; currently preparing for Round 3 – full system autonomy phase.

Organisations

Aero and Robotics Club, BIT Sindri *Jul 2025 – Present*

- Engaged in electronics division, supporting UAV and robotics system development.

Certificates

- Overview of Global Navigation Satellite System (GNSS) – Indian Space Research Organisation (ISRO)
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI (*Coursera*)